Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________ 12. ________ 13. ________
14. ________ 15. ________

Warm-Up: Identify a

Example (Warm-Up: Identify a):
For $y = 4(x-1)^2 + 2$, $a = 4$.

Directions: State the value of a for each equation and whether the parabola opens up or down.

1. $y = 3(x-1)^2 + 2$

3. $y = \frac{1}{4}(x-2)^2 - 3$

2. $y = -2(x+5)^2 + 1$

4. $y = -x^2 + 6$

Identify the Vertex

Example (Identify the Vertex):
For $y = -2(x+4)^2 + 7$, rewrite $(x+4)$ as $(x-(-4))$, so the vertex is $(-4, 7)$.

Directions: Identify the vertex (h, k) for each equation. Show the rewritten form when needed.

5. $y = (x-5)^2 + 3$

8. $y = 5(x+7)^2$

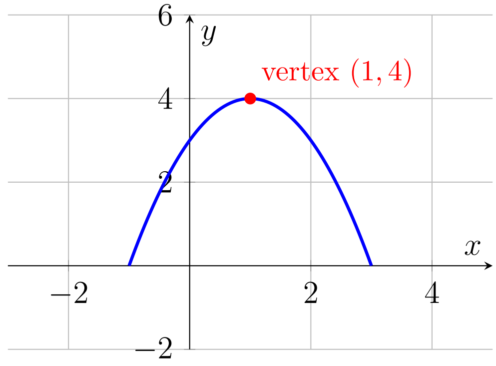
6. $y = 2(x+2)^2 - 6$

9. $y = -4(x-9)^2 - 1$

7. $y = -\frac{1}{2}(x-0)^2 + 8$

Full Identification

Example (Full Identification):
For $y = -3(x-2)^2 + 5$: $a = -3$ (opens down, narrower), vertex $(2, 5)$.



The graph of $y = -(x-1)^2 + 4$.

Directions: For each equation, state a, the vertex, and the direction of opening.

10. $y = -(x-1)^2 + 4$ (use the graph above to confirm your answer)

11. $y = 2(x+3)^2 - 5$

12. $y = -\dfrac{1}{3}(x-6)^2 + 1$

13. $y = 6(x+1)^2 + 2$

Challenge (same sheet):

14. Write a vertex-form equation whose parabola opens down with vertex $(-3, 5)$ and is narrower than $y = x^2$.

15. Two parabolas both have vertex $(2, -1)$, but one has $a = 4$ and the other has $a = \frac{1}{4}$. Explain how their graphs would differ.

Answer Key — fully worked

Procedure: Compare the equation to the pattern $y = a(x - h)^2 + k$.; Read off a – the number multiplied on the squared term.; Rewrite $(x \pm h)$ as $(x - h)$ if needed to correctly identify the sign of h.; Read off k – the constant added or subtracted at the end.; Write the vertex as the ordered pair (h, k).; State the direction of opening: up if $a > 0$, down if $a < 0$..

1. $a = 3$, opens up.

2. $a = -2$, opens down.

3. $a = \frac{1}{4}$, opens up.

4. $a = -1$, opens down.

5. Vertex $(5, 3)$.

6. Vertex $(-2, -6)$.

7. Vertex $(0, 8)$.

8. Vertex $(-7, 0)$.

9. Vertex $(9, -1)$.

10. $a = -1$, vertex $(1, 4)$, opens down – matches the graph shown.

11. $a = 2$, vertex $(-3, -5)$, opens up.

12. $a = -\frac{1}{3}$, vertex $(6, 1)$, opens down.

13. $a = 6$, vertex $(-1, 2)$, opens up.

14. Vertex $(-3, 5)$ means $h = -3$, $k = 5$; opens down means $a < 0$; narrower than $y = x^2$ means $|a| > 1$. One valid answer: $y = -2(x + 3)^2 + 5$ (any coefficient with $a < -1$ works, e.g. $y = -3(x + 3)^2 + 5$).

15. Both share vertex $(2, -1)$ and open up, but $a = 4$ makes the first parabola narrower (a vertical stretch, steeper), while $a = \frac{1}{4}$ makes the second wider (a vertical compression, flatter). At the same horizontal distance from the axis, the $a = 4$ graph rises 16 times higher than the $a = \frac{1}{4}$ graph.